

Supplementary Materials for Submission 4988

Anonymous submission

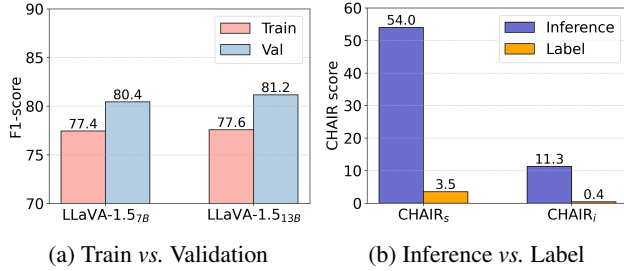


Figure 1: Hallucination analysis in LLaVA-1.5: (a) Performance on seen vs. unseen images, and (b) Regenerated captions vs. original training labels.

More Details on Data Collection Process

To avoid trivial masks on unimportant objects or aggressive masks that remove too much visual information, we constrain the size of the target object to 5%–25% of the image size. If multiple objects satisfy the criteria, we select the least frequent one in the collected data to ensure a balanced distribution of object classes. Once the target object is identified, it is replaced by a black mask using the bounding box provided in the MSCOCO annotations. Finally, we employ human annotators to verify the quality of counterfactual images, ensuring that the target object is no longer visible.

Additional Experiments on the Influence of Training Data on Hallucination

We conduct two additional experiments to further examine how training data influences hallucination. First, using the same pipeline as POPEv2, we collect new data from the MSCOCO 2017 validation set and evaluate LLaVA-1.5. As shown in Figure 1a, the model achieves a higher F1-score on these previously unseen images. Next, we filter caption instructions from LLaVA-1.5’s training data and let the model to regenerate captions on these data. Using the CHAIR metric (Rohrbach et al. 2018), we compare hallucination levels in ground-truth labels and inference results. As illustrated in Figure 1b, while the training labels are nearly hallucination-free, LLaVA-1.5 produces captions with significant hallucinations, even has encountered these images during training. These findings suggest that current LVLMs commonly suffer from

Config	POPEv2		Object HalBench	
	F1 Score	TNR	Resp.	Mention
2k-8k, $\alpha = 0.01$	84.54	70.80	43.0	22.5
2k-8k, $\alpha = 0.02$	87.38	82.20	38.0	19.8
2k-8k, $\alpha = 0.05$	83.57	66.80	49.8	24.9

Table 1: Impact of different unlearning factor α .

training bias, which may stem from the overfitting to specific patterns in the training data. This bias could lead the model to rely excessively on learned priors rather than accurately interpreting the visual input, thereby increasing hallucination rates even on familiar data.

Implementation Details for Probing Experiment

For the data preparation., we select “dog” as the target object for probing and take MSCOCO 2017 as our image source. We first collect 7,000 images with half of the images containing “dog” and the other half not. We then split images into train and validation sets at a ratio of 7:1. For the test set, our goal is to evaluate whether the probes trained on standard images can accurately determine the absence of an object in counterfactual images. Therefore, we construct another 177 counterfactual images by masking the dog contained in each of them. We select LLaVA-1.5-7B as the target model and set the I as the images from our training set and T as “Is there a dog in the image?”. We then extract the encoded image features E_I , the image embeddings H_I^0 , and the image and text hidden states from layers 1, 16, and 32 of the model. These features are used to train the linear probe, and we select the checkpoint according to the performance on validation set.

Ablation on Unlearning Factor α

We also analyze the effect of the unlearning parameter α (set to 0.01, 0.02, and 0.05) on the performance of LLaVA-1.5-7B. Results in Table 1 show that even a small α (0.01) effectively reduces hallucinations. Increasing α to 0.02 further mitigates them, but at $\alpha = 0.05$, performance degrades, with occasional repetitive outputs. A possible reason is that

Table 2: Impact of different data ratios.

Config	POPEv2		Object HalBench	
	F1-score	TNR	Resp.	Mention
2k-2k	86.49	87.40	36.3	20.3
2k-8k	87.38	82.20	38.0	19.8
2k-16k	85.83	88.00	42.7	21.8

the excessive unlearning process would disrupt the model’s original generative capabilities, underscoring the need for careful α calibration.

Ablation on Data Ratio

Finally, we investigate the impact of the ratio between caption data and general visual instruction data (for gradient descending only) on the performance of LLaVA-1.5-7B. We evaluate three distinct data configurations (*i.e.*, 2k-2k, 2k-8k, and 2k-16k mixtures of caption and general instruction data) while training the models for the same number of steps. As shown in Table 2, Obliviate demonstrates strong robustness, effectively mitigating hallucinations across all data ratios. Furthermore, we observe that as the proportion of caption data used for unlearning increases, Obliviate becomes increasingly effective in reducing generative hallucinations. This finding further validates the efficacy of the bias unlearning strategy, which can work well using less data.

Related Work

Large Vision-Language Models. LVLMS (Liu et al. 2024b; Yao et al. 2024; Li et al. 2024) have demonstrated strong performance in multimodal tasks by integrating images into LLMs’ embedding space using a visual encoder and projector. Recent advancements have enhanced their perceptual and cognitive abilities through various techniques. For instance, LLaVA-NeXT (Liu et al. 2024c) processes images at dynamic resolutions by dividing them into grids, while Qwen2-VL (Wang et al. 2024) improves positional encoding for image and video inputs with M-RoPE. LVLMS can also handle both image and video modalities by unifying their processing and instruction mixing (Li et al. 2024; Chen et al. 2024b). Additionally, models like QvQ (Qwen Team 2024) and Virgo (Du et al. 2025) have made significant progress in complex reasoning tasks by leveraging extended chain-of-thought reasoning.

Hallucination Mitigation for LVLMS. Despite recent advancements, hallucination remains a significant challenge for LVLMS, impeding their accurate comprehension of visual inputs (Wang et al. 2023b,a; Chen et al. 2024a). Prior studies have explored various approaches to mitigate hallucinations in LVLMS (Pi et al. 2024; Zhou et al. 2024; Zhao et al. 2023; Yue, Zhang, and Jin 2024; Zhao et al. 2024; Liu et al. 2024a,e; Xing et al. 2024). LLaVA-RLHF (Sun et al. 2024) and RLHF-V (Yu et al. 2024) integrate preference learning techniques such as PPO (Schulman et al. 2017) and DPO (Rafailov et al. 2023) into the LVLMS training process, aligning model

outputs with human to reduce hallucinations. In addition, several works have proposed training-free methods to enhance the decoding stage of LVLMS. VCD (Leng et al. 2024) and VDD (Zhang et al. 2024b) mitigate the influence of language priors by comparing the model’s original outputs to those generated with perturbed visual input. OPERA (Huang et al. 2024) identifies a correlation between hallucinations and knowledge aggregation patterns in the self-attention matrix and introduces a rollback mechanism during decoding to prevent the generation of summary tokens.

Machine Unlearning. Machine unlearning, initially developed for privacy protection, enables models to forget specific data without full retraining. It has been widely applied in various domains like image classification (Ginart et al. 2019), image generation (Gandikota et al. 2023), and LLMs (Liu et al. 2024d). Typical unlearning methods in LLMs involve directly training LLMs on specific dataset to erase knowledge, such as Gradient Ascent (Yao, Xu, and Liu 2024) and NPO (Zhang et al. 2024a). By designing special training objectives (*e.g.*, loss maximization), unlearning methods are more efficient solutions to reduce the memorization of specific data (Zhang et al. 2024a; Liu et al. 2024d). In this work, we apply machine unlearning techniques on LVLMS for hallucination mitigation by employing gradient ascent.

References

- Chen, X.; Ma, Z.; Zhang, X.; Xu, S.; Qian, S.; Yang, J.; Fouhey, D.; and Chai, J. 2024a. Multi-Object Hallucination in Vision Language Models. In *NeurIPS*.
- Chen, Z.; Wang, W.; Tian, H.; Ye, S.; Gao, Z.; Cui, E.; Tong, W.; Hu, K.; Luo, J.; Ma, Z.; Ma, J.; Wang, J.; Dong, X.; Yan, H.; Guo, H.; He, C.; Shi, B.; Jin, Z.; Xu, C.; Wang, B.; Wei, X.; Li, W.; Zhang, W.; Zhang, B.; Cai, P.; Wen, L.; Yan, X.; Dou, M.; Lu, L.; Zhu, X.; Lu, T.; Lin, D.; Qiao, Y.; Dai, J.; and Wang, W. 2024b. How Far Are We to GPT-4V? Closing the Gap to Commercial Multimodal Models with Open-Source Suites. *CoRR*, abs/2404.16821.
- Du, Y.; Liu, Z.; Li, Y.; Zhao, W. X.; Huo, Y.; Wang, B.; Chen, W.; Liu, Z.; Wang, Z.; and Wen, J.-R. 2025. Virgo: A Preliminary Exploration on Reproducing o1-like MLLM. *CoRR*, abs/2501.01904.
- Gandikota, R.; Materzynska, J.; Fiotto-Kaufman, J.; and Bau, D. 2023. Erasing Concepts from Diffusion Models. In *ICCV*, 2426–2436. IEEE.
- Ginart, A.; Guan, M. Y.; Valiant, G.; and Zou, J. 2019. Making AI Forget You: Data Deletion in Machine Learning. In *NeurIPS*, 3513–3526.
- Huang, Q.; Dong, X.; Zhang, P.; Wang, B.; He, C.; Wang, J.; Lin, D.; Zhang, W.; and Yu, N. 2024. OPERA: Alleviating Hallucination in Multi-Modal Large Language Models via Over-Trust Penalty and Retrospection-Allocation. In *CVPR*, 13418–13427. IEEE.
- Leng, S.; Zhang, H.; Chen, G.; Li, X.; Lu, S.; Miao, C.; and Bing, L. 2024. Mitigating Object Hallucinations in Large Vision-Language Models through Visual Contrastive Decoding. In *CVPR*, 13872–13882. IEEE.

- Li, B.; Zhang, Y.; Guo, D.; Zhang, R.; Li, F.; Zhang, H.; Zhang, K.; Li, Y.; Liu, Z.; and Li, C. 2024. LLaVA-OneVision: Easy Visual Task Transfer. *CoRR*, abs/2408.03326.
- Liu, F.; Lin, K.; Li, L.; Wang, J.; Yacoob, Y.; and Wang, L. 2024a. Mitigating Hallucination in Large Multi-Modal Models via Robust Instruction Tuning. In *ICLR*. OpenReview.net.
- Liu, H.; Li, C.; Li, Y.; and Lee, Y. J. 2024b. Improved Baselines with Visual Instruction Tuning. In *CVPR*, 26286–26296. IEEE.
- Liu, H.; Li, C.; Li, Y.; Li, B.; Zhang, Y.; Shen, S.; and Lee, Y. J. 2024c. LLaVA-NeXT: Improved reasoning, OCR, and world knowledge.
- Liu, S.; Yao, Y.; Jia, J.; Casper, S.; Baracaldo, N.; Hase, P.; Xu, X.; Yao, Y.; Li, H.; Varshney, K. R.; Bansal, M.; Koyejo, S.; and Liu, Y. 2024d. Rethinking Machine Unlearning for Large Language Models. *CoRR*, abs/2402.08787.
- Liu, S.; Ye, H.; Xing, L.; and Zou, J. 2024e. Reducing hallucinations in vision-language models via latent space steering. *arXiv preprint arXiv:2410.15778*.
- Pi, R.; Han, T.; Xiong, W.; Zhang, J.; Liu, R.; Pan, R.; and Zhang, T. 2024. Strengthening multimodal large language model with bootstrapped preference optimization. In *European Conference on Computer Vision*, 382–398. Springer.
- Qwen Team. 2024. QVQ: To See the World with Wisdom.
- Rafailov, R.; Sharma, A.; Mitchell, E.; Manning, C. D.; Ermon, S.; and Finn, C. 2023. Direct Preference Optimization: Your Language Model is Secretly a Reward Model. In *NeurIPS*.
- Rohrbach, A.; Hendricks, L. A.; Burns, K.; Darrell, T.; and Saenko, K. 2018. Object Hallucination in Image Captioning. In *EMNLP*, 4035–4045. Association for Computational Linguistics.
- Schulman, J.; Wolski, F.; Dhariwal, P.; Radford, A.; and Klimov, O. 2017. Proximal Policy Optimization Algorithms. *CoRR*, abs/1707.06347.
- Sun, Z.; Shen, S.; Cao, S.; Liu, H.; Li, C.; Shen, Y.; Gan, C.; Gui, L.; Wang, Y.; Yang, Y.; Keutzer, K.; and Darrell, T. 2024. Aligning Large Multimodal Models with Factually Augmented RLHF. In *ACL (Findings)*, 13088–13110. Association for Computational Linguistics.
- Wang, J.; Wang, Y.; Xu, G.; Zhang, J.; Gu, Y.; Jia, H.; Wang, J.; Xu, H.; Yan, M.; Zhang, J.; et al. 2023a. Amber: An llm-free multi-dimensional benchmark for mllms hallucination evaluation. *arXiv preprint arXiv:2311.07397*.
- Wang, J.; Zhou, Y.; Xu, G.; Shi, P.; Zhao, C.; Xu, H.; Ye, Q.; Yan, M.; Zhang, J.; Zhu, J.; et al. 2023b. Evaluation and analysis of hallucination in large vision-language models. *arXiv preprint arXiv:2308.15126*.
- Wang, P.; Bai, S.; Tan, S.; Wang, S.; Fan, Z.; Bai, J.; Chen, K.; Liu, X.; Wang, J.; Ge, W.; Fan, Y.; Dang, K.; Du, M.; Ren, X.; Men, R.; Liu, D.; Zhou, C.; Zhou, J.; and Lin, J. 2024. Qwen2-VL: Enhancing Vision-Language Model’s Perception of the World at Any Resolution. *CoRR*, abs/2409.12191.
- Xing, S.; Zhao, F.; Wu, Z.; An, T.; Chen, W.; Li, C.; Zhang, J.; and Dai, X. 2024. EFUF: Efficient Fine-Grained Unlearning Framework for Mitigating Hallucinations in Multimodal Large Language Models. In *EMNLP*, 1167–1181. Association for Computational Linguistics.
- Yao, Y.; Xu, X.; and Liu, Y. 2024. Large Language Model Unlearning. In *NeurIPS*.
- Yao, Y.; Yu, T.; Zhang, A.; Wang, C.; Cui, J.; Zhu, H.; Cai, T.; Li, H.; Zhao, W.; He, Z.; Chen, Q.; Zhou, H.; Zou, Z.; Zhang, H.; Hu, S.; Zheng, Z.; Zhou, J.; Cai, J.; Han, X.; Zeng, G.; Li, D.; Liu, Z.; and Sun, M. 2024. MiniCPM-V: A GPT-4V Level MLLM on Your Phone. *CoRR*, abs/2408.01800.
- Yu, T.; Yao, Y.; Zhang, H.; He, T.; Han, Y.; Cui, G.; Hu, J.; Liu, Z.; Zheng, H.; and Sun, M. 2024. RLHF-V: Towards Trustworthy MLLMs via Behavior Alignment from Fine-Grained Correctional Human Feedback. In *CVPR*, 13807–13816. IEEE.
- Yue, Z.; Zhang, L.; and Jin, Q. 2024. Less is More: Mitigating Multimodal Hallucination from an EOS Decision Perspective. In *ACL (I)*, 11766–11781. Association for Computational Linguistics.
- Zhang, R.; Lin, L.; Bai, Y.; and Mei, S. 2024a. Negative Preference Optimization: From Catastrophic Collapse to Effective Unlearning. *CoRR*, abs/2404.05868.
- Zhang, Y.; Yu, W.; Wen, Q.; Wang, X.; Zhang, Z.; Wang, L.; Jin, R.; and Tan, T. 2024b. Debiasing Multimodal Large Language Models. *CoRR*, abs/2403.05262.
- Zhao, H.; Si, S.; Chen, L.; Zhang, Y.; Sun, M.; Zhang, M.; and Chang, B. 2024. Looking Beyond Text: Reducing Language bias in Large Vision-Language Models via Multimodal Dual-Attention and Soft-Image Guidance. *CoRR*, abs/2411.14279.
- Zhao, Z.; Wang, B.; Ouyang, L.; Dong, X.; Wang, J.; and He, C. 2023. Beyond Hallucinations: Enhancing LVLMs through Hallucination-Aware Direct Preference Optimization. *CoRR*, abs/2311.16839.
- Zhou, Y.; Cui, C.; Rafailov, R.; Finn, C.; and Yao, H. 2024. Aligning Modalities in Vision Large Language Models via Preference Fine-tuning. *CoRR*, abs/2402.11411.